MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, ADJUSTABLE POSITIVE VOLTAGE REGULATORS, MONOLITHIC, SILICON

This amendment forms a part of MIL-M-38510/117A, dated 11 June 1981, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, Scope: Delete "Three" product assurance classes, and substitute, "Two" product assurance classes.

PAGE 2

1.4: Delete "Case" and substitute "Ambient".

PAGES 3 THROUGH 8

TABLE I: Delete all references in table to case temperature "T_C", and substitute "T_A" ambient temperature.

TABLE I, V_{OUT}(RECOV), load current column: Delete "C_L = 0.1 µF" as substituted in amendment 1 and reinstate "C_L = 20 µF". Make this same change for all device types on pages 4, 5, 6, 7, and 8.

TABLE I, V_{START}: Delete "C_L = 0.1 µF" as substituted in amendment 1 and reinstate "C_L = 20 µF". Make this same change for all devices types on pages 4, 5, 6, 7, and 8.

PAGE 5

TABLE I, Load regulation, V_{RLOAD}, limits column: Delete "±3.5 mV" and substitute "±12 mV".

TABLE I, Thermal regulation, (V_{RTH}), load current column: Delete "I_L = -750 mA" and substitute "I_L = -500 mA". Also, in the limits column delete "±5 mV" and substitute "±12 mV".

PAGE 6

TABLE I, Load regulation (V_{RLOAD}), load current column: Delete "150mA $\leq I_L \leq$ -5 mA" and substitute "-200 mA $\leq I_L \leq$ -5 mA".

TABLE I, Thermal regulation (V_{RTH}), limits column: Delete "±5 mV" and substitute "±12 mV".

TABLE I, I_Q, maximum limits column: Delete "-0.50" and substitute "-0.20".

*The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
21	21
22	22

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PAGE 9

3.4: Delete "case temperature" and substitute "ambient temperature".

3.5, delete and substitute the following: "The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III."

3.6, delete and substitute the following:

"3.6 Marking. Marking shall be as specified in MIL-M-38510."

PAGE 10

TABLE II, Electrical test requirements: Delete class C devices column in its entirety.

4.2.e, delete and substitute the following: "Percent defective allowable (PDA) - The PDA for class S and class B devices shall be as specified in MIL-M-38510."

PAGE 26

Note 2, line 2: Delete "shall" and substitute "should" and add the following: "For device type 03 only. If output voltage measurements are not made at the case but instead at the output lead, an error will result in the measurement due to internal lead resistance. The amount of error depends on the magnitude of the load current and the distance from the case to where the output voltage measurement is taken on the output pin."

Add note: "16/ For static test V_{RLOAD} 1, I_L = 500 mA, device type 03 only the following may apply. If output voltage measurements are taken from the output lead and not the case, the maximum limit shall be allowed to increase by 5 mV to account for the error due to internal lead resistance."

PAGE 36

TABLE III: Delete "T_C" and substitute "T_A" throughout entire table III.

PAGE 42

TABLE III, test no. 6, limits column: Delete "±3.5 mV".

TABLE III, test no. 7, limits column: Add "±12 mV".

TABLE III, test no. 8, limits column: Delete "±3.5 mV" and substitute "±12 mV".

TABLE III, test no. 9: Delete "I_L = -750 mA" and in the limits column "±5 mV" and substitute "I_L = -500 mA" and "±12 mV", respectively.

PAGE 45

TABLE III, test no. 9, limits column: Delete "±5 mV" and substitute "±12 mV".

PAGES 45 THROUGH 47

TABLE I, I_O, maximum limits column, test nos. 19 and 20; 40 and 41; and 62 and 63, respectively: Delete "-0.50" and substitute "-0.20".

PAGE 48

TABLE III, test no. 6, limits column: Delete "±3.5 mV".

TABLE III, test no. 7, limits column: Add "±3.5 mV".

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PAGE 51

TABLE III, test no. 7, limits column: Delete "±3.8 mV".

TABLE III, test no. 8, limits column: Add "±3.8 mV".

PAGE 54

4.4.3a and 4.4.3b: Delete reference to class C devices (2 places).

4.5, delete and substitute the following: "Methods of inspection. Methods of inspection shall be as specified in MIL-M-38510."

PAGE 55

TABLE IV, symbol V_{OUT}, for all device types: Add 2/ to each V_{OUT} listed; also following footnote "1/" add "2/ Delta limits apply to test no. 3 for all device types."

TABLE IV, symbol I_{ADJ} for device types 03, 04, 05, 06: Add "3/" to the I_{ADJ} listed; also following footnotes "1/" and "2/" add "3/ Delta limits apply to test number 11 for 03, 04, 05 device types and test number 12 for device type 06."

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liabilities whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Air Force - 17
Army - ER
Navy - EC
NASA - NA

Review activities:
Air Force - 11, 19, 85, 99
Army - AR, MI
Navy - OS, SH
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

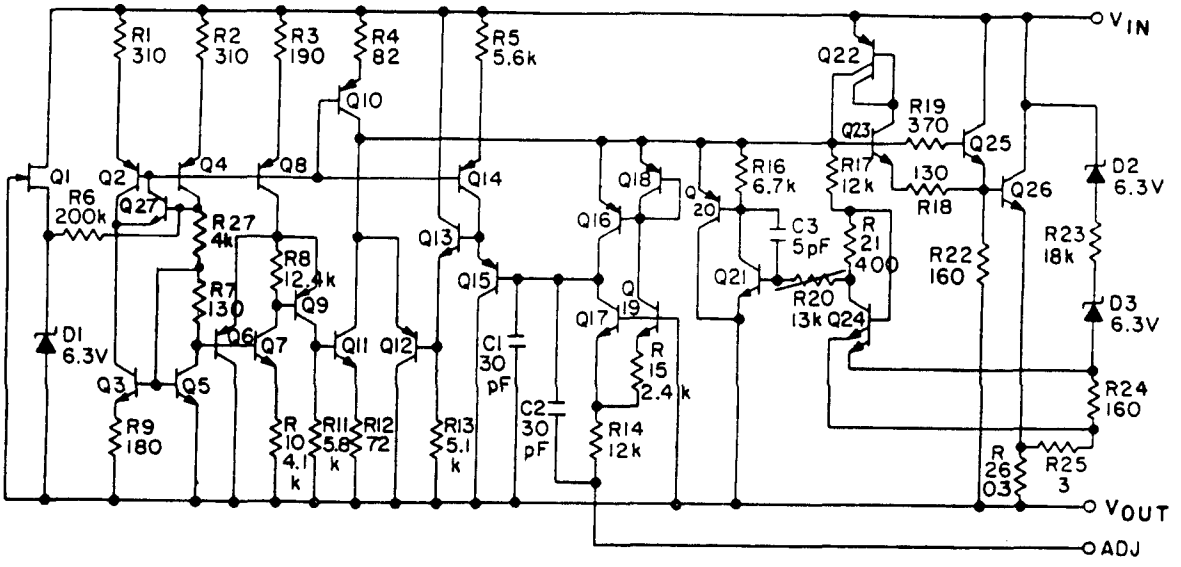
Preparing activity:
Air Force - 17

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(Project 5962-0991)

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Device type 05



Device type 06

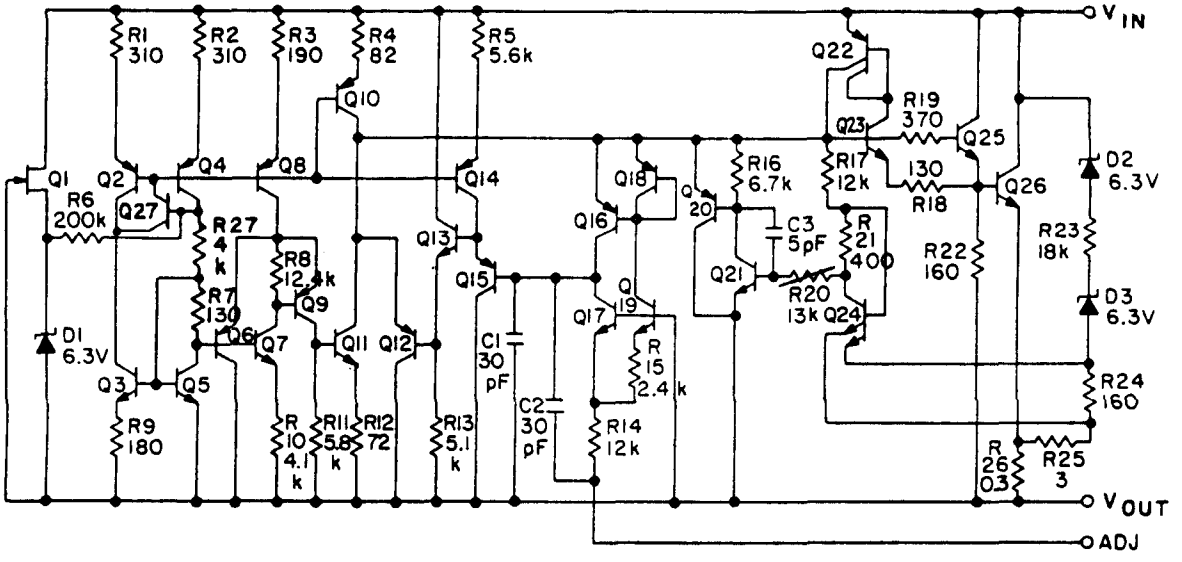
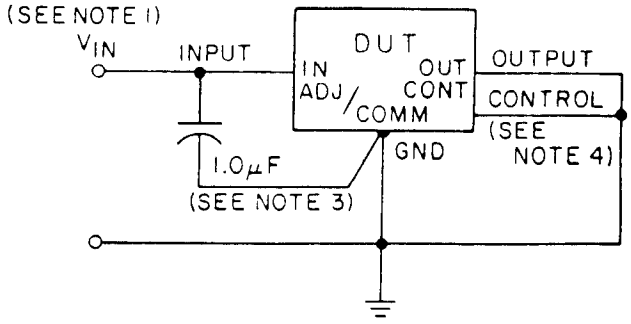


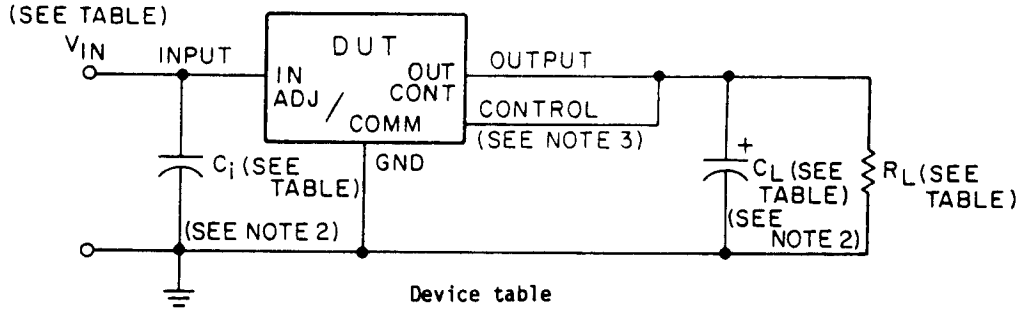
FIGURE 9. Schematic circuit - Continued.

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NOTES:
1. For device types 01, 02, 03 and 04 V_{IN} = 40 V (0% to +10% tolerance).
   For device types 05 and 06 V_{IN} = 35 V (-5% to 0% tolerance).
2. Test is conducted at T_A = 25°C without a heat sink for a minimum of 30 minutes.
3. The 1.0 μF capacitor may be reduced or eliminated provided the device remains stable.
4. The control pin connection is required for device types 01 and 02 only.

FIGURE 10. Pre burn-in test circuit.



Device table

Device type	01	02	03	04	05	06
V_{IN}	20 V	20 V	36.5 V	36.5 V	36.5 V	36.5 V
R_L	887Ω	118Ω	250Ω	63.4Ω	63.4Ω	63.4Ω
C_I	0.33 μF	0.33 μF	1.0 μF	1.0 μF	1.0 μF	1.0 μF
C_L	0.1 μF	0.1 μF	1.0 μF	1.0 μF	1.0 μF	1.0 μF

NOTES:
1. Test is conducted without a heat sink.
2. The capacitors may be reduced or eliminated provided the device remains stable.
3. The control pin connection is required for device types 01 and 02 only.

FIGURE 11. Burn-in and operating life test circuit.

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